

A REDUCTION, A CLASSIFICATION, AND A VERIFICATION UP TO 10^{12} FOR ERDŐS PROBLEM #458

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ABSTRACT. Erdős and Graham asked whether $[1, \dots, p_{k+1}-1] < p_k [1, \dots, p_k]$ for all $k \geq 1$, where $[1, \dots, n] = \text{lcm}(1, \dots, n)$ and p_k is the k -th prime (Erdős problem #458). We reduce the question to the cohabitation of prime powers inside a single prime gap, classify rigorously all configurations that could produce a counterexample, and verify the inequality for all k with $p_{k+1} \leq 10^{12}$ by an exhaustive enumeration of the 80 070 proper prime powers below that bound: exactly five prime gaps below 10^{12} contain more than one prime power, the largest ratio being $6/7$ at the gap $(7, 11)$. As a consequence, the inequality holds for all sufficiently large k under a Legendre-type gap hypothesis together with a weak Pillai-type hypothesis on near-coincidences of prime powers. Every statement below is labelled as rigorous, conditional, or heuristic.

1. THE PROBLEM

Throughout, $[1, \dots, n]$ denotes $\text{lcm}(1, 2, \dots, n)$ and p_k the k -th prime. Erdős and Graham [3, p. 91] asked:

Is it true that, for all $k \geq 1$,

$$(1) \quad [1, \dots, p_{k+1} - 1] < p_k [1, \dots, p_k]?$$

This is problem #458 in Bloom's catalogue [2], where it is recorded as open and falsifiable; the related OEIS sequence is A056604 [6]. Erdős and Graham write that (1) is “almost certainly” true but that a proof is beyond reach, the first obstruction being the possibility of several primes q with $p_k < q^2 < p_{k+1}$, whose exclusion is essentially Legendre's conjecture; they add that “the small primes also cause trouble.” The purpose of this note is to make both obstructions precise and exhaustive: we classify *all* configurations that could violate (1) (Theorem 2.4), we determine *exactly* which small primes cause trouble below 10^{12} (Section 3), and we record the resulting conditional statement (Theorem 4.3).

2. REDUCTION AND CLASSIFICATION

[Rigorous.] The following facts are elementary; we include the short proofs for completeness.

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Lemma 2.1. *For $n \geq 2$, $[1, \dots, n]/[1, \dots, n-1] = q$ if $n = q^a$ is a prime power ($a \geq 1$), and $= 1$ otherwise.*

Proof. Standard: $[1, \dots, n] = \exp(\psi(n))$ with ψ the second Chebyshev function, which jumps by $\log q$ exactly at prime powers q^a . $\square$

Fix $k \geq 1$ and write $p = p_k$, $P = p_{k+1}$. Since the open interval (p, P) contains no prime, Lemma 2.1 gives

$$(2) \quad \frac{[1, \dots, P-1]}{[1, \dots, p]} = \prod_{\substack{q^a \in (p, P) \\ a \geq 2}} q =: R_k,$$

the product running over the proper prime powers lying strictly between p and P (equivalently, in $(p, P-1]$). Hence:

Lemma 2.2. *Inequality (1) holds at k if and only if $R_k < p_k$.*

Lemma 2.3. *Let $p \geq 2$ and let q^a, r^b ($a, b \geq 2$) be proper prime powers in (p, P) . Then:*

- (i) *distinct prime powers in (p, P) have distinct bases;*
- (ii) *if (p, P) contains exactly one proper prime power q^a , then $R_k = q < p$;*
- (iii) *if (p, P) contains exactly two proper prime powers with exponent multiset $\{2, a\}$, $a \geq 3$, then $R_k < p$ provided $p > 32$;*
- (iv) *if (p, P) contains exactly two proper prime powers with both exponents ≥ 3 , then $R_k < p$ provided $p > 4$.*

Proof. By Bertrand's postulate, $P < 2p$, so every prime power in (p, P) lies in $(p, 2p)$. (i) If $q^a < q^b$ lie in $(p, 2p)$ then $q^b \geq q \cdot q^a \geq 2q^a > 2p$, a contradiction. (ii) $q \leq (q^a)^{1/2} < (2p)^{1/2} < p$ for all $p \geq 2$. (iii) $R_k = qr < (2p)^{1/2}(2p)^{1/3} = (2p)^{5/6}$, and $(2p)^{5/6} < p \iff p > 2^5 = 32$. (iv) $R_k = qr < (2p)^{2/3}$, and $(2p)^{2/3} < p \iff p > 4$. $\square$

Theorem 2.4 (Classification of potential violations). *Let $p_k \geq 37$. If (1) fails at k , then the gap (p_k, p_{k+1}) contains*

- (I) *two squares of distinct primes, $q^2 < r^2$; or*
- (II) *two proper prime powers $q^a < r^b$ with $a, b \geq 3$.*

Moreover, in case (I) the failure is automatic: two distinct prime squares in one gap always force $R_k > p_k$. In case (II) with $a = b$, the gap satisfies $p_{k+1} - p_k > \frac{3}{2^{1/3}} p_k^{2/3}$.

Proof. Suppose $R_k \geq p_k$ and let m be the number of proper prime powers in the gap. By Lemma 2.3(ii)–(iv), $m \leq 2$ forces either $m = 2$ with both exponents equal to 2 — case (I) — or no failure. If $m \geq 3$ and the gap does not contain two prime squares, then at most one exponent equals 2, so at least two exponents are ≥ 3 — case (II).

Automatic failure in (I): $q^2 > p_k$ gives $q > \sqrt{p_k}$, and $r > q$ gives $R_k \geq qr > q^2 > p_k$.

Gap bound in (II) with $a = b \geq 3$: the bases are distinct primes by Lemma 2.3(i), so $r \geq q + 1$ and $r^a - q^a \geq a q^{a-1} = a q^a / q > 3 p_k / (2 p_k)^{1/3} = \frac{3}{2^{1/3}} p_k^{2/3}$, using $q \leq (2 p_k)^{1/a} \leq (2 p_k)^{1/3}$. $\square$

Remark 2.5. [Rigorous.] Case (I) forces a prime gap longer than $2\sqrt{p_k}$: indeed $r \geq q + 1$ gives $p_{k+1} - p_k > r^2 - q^2 \geq 2q + 1 > 2\sqrt{p_k}$. Note that even the Riemann Hypothesis only yields gaps $O(\sqrt{p} \log p)$, which does not exclude case (I); its exclusion is of the strength of Legendre's conjecture, as already observed in [2]. Two distinct prime squares $q^2 < r^2$ in one gap would in particular leave the interval $(q^2, (q+1)^2)$ free of primes, i.e. exhibit a counterexample to Legendre's conjecture.

Remark 2.6. [Rigorous / conditional.] In case (II) with equal exponents, the gap exceeds $\frac{3}{2^{1/3}} p_k^{2/3}$, which contradicts $p_{k+1} - p_k \ll p_k^{0.525}$ [1] for all sufficiently large k ; below any fixed bound the configuration is checked directly (Section 3). In case (II) with distinct exponents $a \neq b$, the two powers satisfy $|q^a - r^b| < p_{k+1} - p_k$ with $q^a, r^b \approx p_k$: a Pillai-type near-coincidence of perfect powers. Such configurations are expected to be extremely rare, but their unconditional exclusion is open; this is a second, less visible obstruction to (1), distinct from the Legendre-type one.

3. EXHAUSTIVE VERIFICATION UP TO 10^{12}

[Rigorous computation.] By Lemma 2.3(ii), a violation requires at least two proper prime powers in one gap. We therefore enumerated *all* proper prime powers $q^a \leq 10^{12}$ ($a \geq 2$) — there are 80 070 of them — sorted them, and for each consecutive pair tested whether a prime lies strictly between them; if not, the pair cohabits in a single prime gap, which we then reconstructed in full and tested via Lemma 2.2. All primality tests concern integers below 2^{64} , where the Baillie–PSW test as implemented in SymPy is deterministic (no BPSW pseudoprime exists below 2^{64} [4]). The script is reproduced verbatim in Appendix A.

The results were cross-checked by three fully independent computations: (a) a brute-force evaluation of the exact integers $[1, \dots, n]$ by iterated gcd, with no recourse to the reduction of Section 2, confirming (1) directly for all $p_{k+1} \leq 10^5$; (b) a segmented Eratosthenes sieve up to 10^9 , using no probabilistic primality test whatsoever, which recovers exactly the same five cohabitations; (c) a re-run of the full enumeration up to 10^{12} using a self-contained deterministic Miller–Rabin test with the witness set $\{2, 3, \dots, 37\}$ (exact for $n < 3.317 \cdot 10^{24}$ [5]), independent of SymPy, with identical output.

Theorem 3.1. *Inequality (1) holds for every k with $p_{k+1} \leq 10^{12}$. Exactly five prime gaps below 10^{12} contain more than one proper prime power, namely:*

Gap (p_k, p_{k+1})	Prime powers	R_k	R_k/p_k
(7, 11)	$2^3, 3^2$	6	0.8571
(23, 29)	$5^2, 3^3$	15	0.6522
(113, 127)	$11^2, 5^3$	55	0.4867
(2179, 2203)	$3^7, 13^3$	39	0.0179
(32749, 32771)	$181^2, 2^{15}$	362	0.0111

In particular, the maximum of R_k/p_k over the whole range is $6/7$, attained at $k = 4$, and every prime gap with $32771 < p_{k+1} \leq 10^{12}$ contains at most one proper prime power.

Remark 3.2. [Rigorous.] No prime gap below 10^{12} contains two distinct prime squares; all five cohabitations above are of mixed type (one square and one higher power, or two higher powers of distinct exponents), and none comes close to a violation. The two configurations of Theorem 2.4 are thus both empirically absent in a wide range, in line with the heuristic of Remark 4.4.

4. A CONDITIONAL STATEMENT

Hypothesis 4.1 (L). There is k_0 such that $p_{k+1} - p_k \leq 2\sqrt{p_k}$ for all $k \geq k_0$.

Hypothesis 4.2 (P). Only finitely many pairs of proper prime powers $q^a \neq r^b$ with $a, b \geq 3$ satisfy $|q^a - r^b| \leq 2\sqrt{\max(q^a, r^b)}$.

Hypothesis (L) follows from Legendre-type conjectures (and from Cramér's model); Hypothesis (P) is a very weak form of Pillai's conjecture. Neither is known.

Theorem 4.3 (Conditional). Assume Hypotheses (L) and (P). Then (1) holds for all sufficiently large k .

Proof. Let k be large and suppose (1) fails. By Theorem 2.4, the gap contains configuration (I) or (II). Configuration (I) forces $p_{k+1} - p_k > 2\sqrt{p_k}$ (Remark 2.5), contradicting (L). In configuration (II), under (L) the two powers satisfy $|q^a - r^b| < 2\sqrt{p_k} \leq 2\sqrt{\max(q^a, r^b)}$; for $a = b$ this contradicts the gap bound of Theorem 2.4 (which exceeds $2\sqrt{p_k}$ for p_k large), and for $a \neq b$ hypothesis (P) leaves only finitely many possibilities. $\square$

Remark 4.4. [Heuristic.] In Cramér's model the expected number of prime gaps ever containing two prime squares is essentially zero ($\sum_q \exp(-c\sqrt{q^2}/\log q^2) < \infty$ -type estimates), and Pillai near-coincidences of size $\sqrt{x}$ near x are likewise expected to be finite in number. The data of Theorem 3.1 — five cohabitations, all below 33 000, with R_k/p_k decaying rapidly — is consistent with this. We emphasise, however, that a proof of (1) appears out of reach: excluding (I) requires gap bounds of strength $\sqrt{p}$, beyond even the Riemann Hypothesis, and excluding (II) for $a \neq b$ requires Pillai-type results that are equally unavailable.

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REFERENCES

- [1] R. C. Baker, G. Harman, J. Pintz, *The difference between consecutive primes, II*, Proc. London Math. Soc. (3) **83** (2001), 532–562.
- [2] T. F. Bloom, *Erdős Problem #458*, <https://www.erdosproblems.com/458>, accessed June 2026.
- [3] P. Erdős, R. L. Graham, *Old and new problems and results in combinatorial number theory*, Monographies de L'Enseignement Mathématique **28**, Genève, 1980, p. 91.
- [4] J. Gilchrist, *Pseudoprime enumeration with probabilistic primality tests* (verification that no Baillie–PSW pseudoprime exists below 2^{64} , building on J. Feitsma's tables), 2013. [Reference to be finalised before posting.]
- [5] J. Sorenson, J. Webster, *Strong pseudoprimes to twelve prime bases*, Math. Comp. **86** (2017), 985–1003. [Bound $3.317 \cdot 10^{24}$ for witnesses $\{2, \dots, 37\}$; reference to be double-checked before posting.]
- [6] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, sequence A056604, <https://oeis.org/A056604>.

APPENDIX A. VERIFICATION SCRIPT (PYTHON 3.12, EXACT ARITHMETIC)

```

from sympy import primerange, isprime, nextprime
from math import isqrt
import bisect

LIMIT = 10**12
powers = []                                # (value, base, exponent), exponent >= 2
for q in primerange(2, isqrt(LIMIT) + 1):
    v, a = q*q, 2
    while v <= LIMIT:
        powers.append((v, q, a)); v *= q; a += 1
powers.sort()                               # 80,070 proper prime powers

cohab = []                                  # adjacent pairs with no prime in between
for i in range(len(powers) - 1):
    v1 = powers[i][0]; v2 = powers[i+1][0]
    if nextprime(v1) > v2:                   # v1, v2 composite => same prime gap
        cohab.append((powers[i], powers[i+1]))

violations = 0
for (v1, q1, a1), (v2, q2, a2) in cohab:
    lo = v1
    while not isprime(lo): lo -= 1           # p_k
    hi = nextprime(v2)                      # p_{k+1}
    iL = bisect.bisect_right(powers, (lo, 10**18, 0))
    iR = bisect.bisect_left(powers, (hi, 0, 0))

```

```
grp = powers[iL:iR]
R = 1
for v, q, a in grp: R *= q
print(lo, hi, grp, R, R/lo)
if R >= lo: violations += 1
print("violations:", violations)          # output: 0
```

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